Perspective and the processing of Chinese reflexives: A visual world eye-tracking study

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Introduction Chinese reflexives *ziji* ('self') and *ta-ziji* ('s/he-self') can be exempt from the locality constraint (e.g., [1]) and be bound by a perspectival center (e.g., [3], [6]). However, two open questions remain. First, it is debated when discourse cues are accessed during processing (e.g., [2], [4]) (**Exp.1**). Second, it is unclear how closely the two Chinese reflexives pattern together during online processing (**Exp.2**). In two visual world eye-tracking studies, we investigated how perspective-related cues – discourse topicality, logophoric role (source vs. perceiver), and 1st-person blocking (i.e., the blocking effect: speakers prioritize other own perspectives, see [3]) – influence online reflexive resolution in Chinese.

To preview the results, we find that perspective cues can guide reflexive resolution since early processing stages. However, different types of discourse cues affect reflexive resolution differently. Furthermore, *ziji* and *ta-ziji* show different sensitivities to 1st-person blocking cues. Together, these results support a form-specific multiple-constraints approach to online reflexive resolution.

Methods Exp. 1 ($N_{participants}$ = 61) examined *ziji*. We crossed *Topicality* (Topical/Neutral) and *Logophoric Role* (Source/Perceiver). Topicality refers to the topical status of the non-local subject, manipulated via context sentences (see (1)). The Logophoric Role refers to whether the non-local subject is a source or a perceiver, manipulated via the matrix verb ('tell' or 'hear'). As discourse-topical and source referents are more likely to serve as perspectival centers than neutral and perceiver referents, respectively ([4], [5]), both factors are expected to increase the probability of the non-local binding of *ziji*.

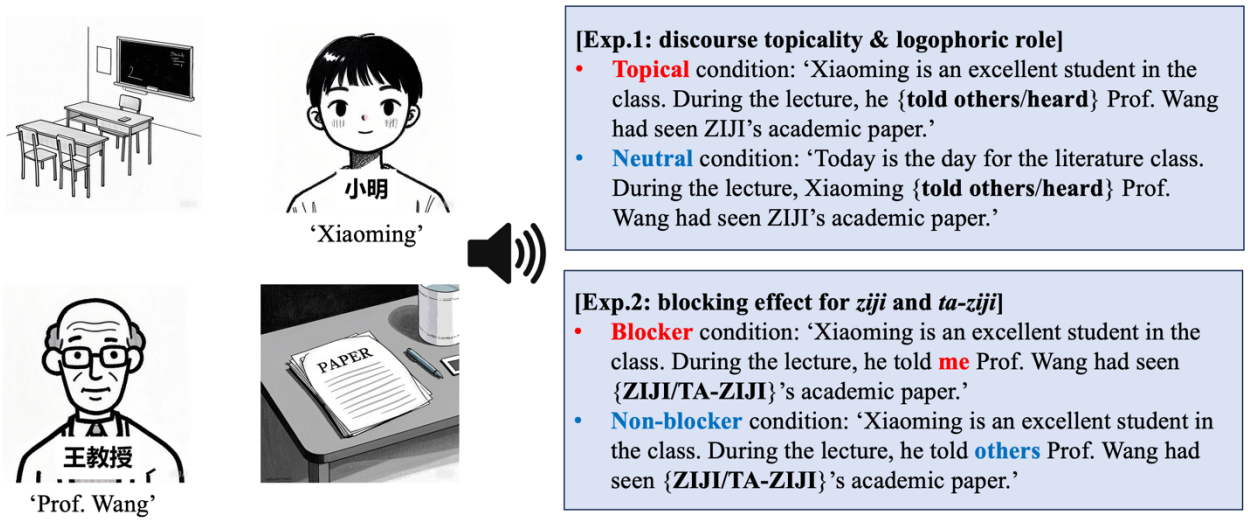
Exp.2 ($N_{participants}$ = 60) compared the processing of *ziji* and *ta-ziji* by crossing the factors *Reflexive* (*ziji/ta-ziji*) and *Blocker Type* (Blocking/Non-blocker) (see (1)). We contrasted the 1st-person blocker ('me') with the 3rd-person non-blocker ('others') to examine how and when the two reflexives show the blocking effect (i.e., increase of local binding). In Exp.2, the non-local subject is always topical, and the 1st-person blocker ('me') is in object position.

For each experiment, twenty-four target picture-audio sets were created, along with 36 fillers. Each picture target set includes the local subject, the non-local subject, and two distractor objects. The 4 pictures were evenly positioned in the 4 corners of the screen throughout the experiment. To minimize recency effects favoring the local subject, a black-box flashed for 50ms at the screen center 250ms before the reflexive onset. Participants clicked on the last-mentioned character (i.e., antecedent) after hearing the audio.

Results Fig. 1 (Exp. 1) and **Fig.2** (Exp. 2) display the results. For Exp.1, we plotted the *Non-local Advantage Proportion (NAP)*, calculated by subtracting the proportion of looks to the non-local subject from that to the local subject; for Exp.2, we plotted the *Local Advantage Proportion (LAP)*, the proportion of looks to the local subject minus that to the non-local subject. The analysis of click responses suggests that (perspective-related) discourse cues modulate antecedent choices as (i) discourse topicality and the source role elicited more non-local choices in Exp.1 ($ps < 0.05$) and (ii) both reflexives showed the blocking effect in Exp.2 ($p < 0.05$). To evaluate when discourse cues influence reflexive resolution, we focused on three 400ms time windows after the onset of the reflexive: 200–600ms, 600–1000ms, and 1000–1400ms. In Exp. 1, the logophoric role effect (source advantage) emerged early during the 200–600ms window ($p < 0.05$), while the discourse topicality effect was delayed until 600–1000ms ($p < 0.05$). In Exp. 2, the 1st-person blocker affected the two reflexives differently: the blocking effect appeared early for *ta-ziji* (200–600ms, $p < 0.05$), but did not emerge for *ziji* until the 600–1000ms window ($p < 0.05$).

Conclusion We have two main findings. First, logophoric cues have an earlier influence on reflexive resolution than discourse topicality, likely due to a recency advantage. Second, the 1st-person (object) blocking effect emerges earlier for *ta-ziji* than for *ziji*, possibly because participants more readily identified with the non-local topical character when processing *ziji*.

(1) An example target set (audio stimuli in Chinese, translated to English below in boxes) in Exp.1 and Exp.2.



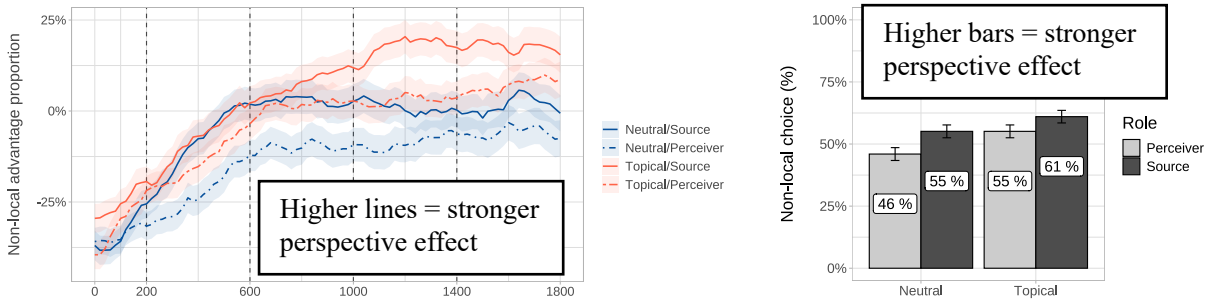


Figure 1: Non-local advantage proportions from the onset of the reflexive *ziji* (left panel) and click responses (right panel) in **Exp.1**.

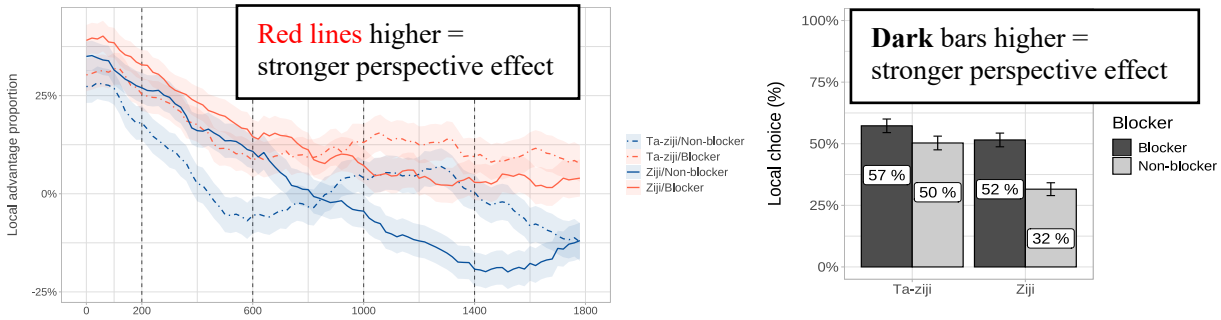


Figure 2: Non-local advantage proportions from the onset of the reflexive (*ta-*)*ziji* (left panel) and click responses (right panel) in **Exp.2**.

References

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